Mutualism

Learning objectives:

1. Understand the costs and benefits of different mutualistic relationships
2. Incorporate positive interactions in a consumer-resource model

density dependence, other plant species, and natural enemies limit population growth

today focusing on processes that would increase population growth

Terminology

- symbiosis: any relationship between species *broad term, can be + or -, includes parasitism*
- mutualism: specific symbiosis, both species benefit
- commensalism: specific symbiosis, 1 benefits, 1 neutral

Facilitation

- Interspecific, nurse plants in arid environment

Draw number of interspecific neighbors on x, individual fecundity on y, multiple lines for multiple stress levels (rain shelter, natural, supplemental rain): stress increasing, favorable decreasing; slopes highlight this change

- Intraspecific, plant-soil feedbacks

could alter previous plot to intraspecific neighbors on x

- Benefits: increased vital rates in favorable environment
- Costs: competition for space or other resources

Pollination

- pollinators get nutrients, plants get fertilization/higher reproduction
- Benefits: quality seeds, genetic variation *outcrossing, compared to self-pollination, wind pollination*
- Costs: dependent on animal density, nectar production, cross-contamination *pollen from other plants not your own*

Mycorrhizal fungi

- Endomycorrhiza: fungi in root cells, arbuscular mycorrhiza *found in many plant taxa, others specialize*
- Ectomycorrhiza: fungi influence root cells but not inside cell, trees and shrubs
- fungi explore efficiently
  - example: extend beyond normal root zone in crops & get farther nutrients
  - drought resistance in tomatoes and peppers
- fungi get carbohydrates, plants get nutrients
- Benefits: get nutrients or water in resource-poor environments
- Costs: increased carbon costs *fungi take carbs from plant, so are carbon sink*

Modeling

- herbivore as pollinator - life stage-dependent
  - example: butterflies & moths
  - larval/caterpillar: herbivore
  - adults: pollinator
- p - pollination rate
- h_p - handling time per flower
- H - insect herbivore/pollinator abundance
- M - mutualism benefit, proportion

$$M = \frac{pH}{1 + ph_pH}$$

- effect on fecundity
- f_μ - mean fecundity
- fecundity increased if pollinators present

$$f_{eff} = f_\mu(1 + M)$$

- affects plant pops, not herbs
- in plant paps, affects new seeds, not seedbank or seeds consumed:

$$P_t = \frac{f_\mu g N_t}{1 + \alpha g N_t}$$